

# Lesson 3





## Study Information and Tips



This Study Text replicates the video content for the lesson



Supplementary knowledge can also be found in the information sheets and text boxes that accompany the videos, as well as in the resources in the Insights section



Activities to help you consolidate your knowledge are deliberately delivered through practice questions and exercises found in the Test and Apply sections on the VLE. These help you to further build on your knowledge and apply the learning in a way that supports your assessment



You can use this Study Text to test your understanding of the learning content at the end of each lesson. It may also be helpful to refer to it when preparing for your assessments



We recommend that you also produce your own notes in addition to the Study Text. Research has shown that handwritten notes can be very effective for learning retention



If you decide to print this Study Text, set the layout to print 4 or 6 slides per page to save on paper and ink



# Technology and the Business

**Role of technology and  
its applications in the  
workplace**

**Using business software  
packages to produce  
documentation, optimise  
day-to-day tasks and  
support communication**

**Key digital skills**

**Future of technology – key  
trends and opportunities**

# ICT and Workplace Software

Information and communication technology (ICT) can be defined as “a diverse set of technological tools and resources used to transmit, store, create, share or exchange information” (UNESCO IIEP Learning Portal).

In the modern workplace, ICT plays a vital role in how organisations operate, communicate and grow. All organisations use a range of ICT software to help run their business operations. The packages will vary from company to company, depending on their needs.

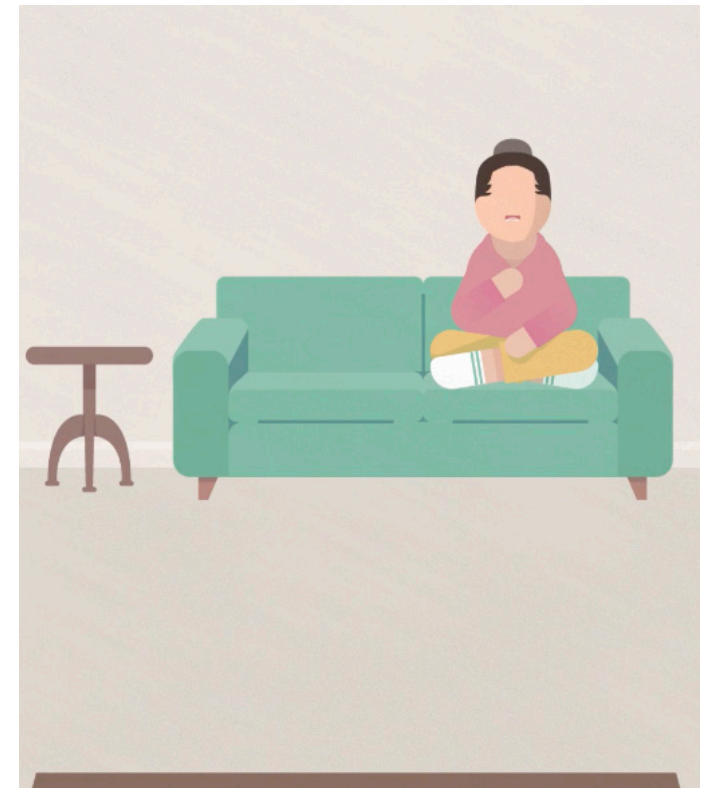




# Role of ICT at Work

To understand the part that ICT plays in business, it's perhaps useful to think about a world without it. From a personal point of view, that might mean no mobile phone, laptop, tablet, or smart TV.

But in business, it would affect every individual piece of information – in different places, at all levels across a company. Data would have to be written on paper, stored by each person, and sent manually.





# Role of ICT at Work (cont.)

This would make it easy to lose information and duplicate effort. And, while we'd still have to share documents, record transactions and make strategic decisions, everything would take much more time. Ultimately, without ICT, every part of working life would be less efficient.

Thankfully, we only have to imagine this situation. Different types of software with a range of roles are used throughout all business processes.

From planning and operations, to managing relationships – software makes organisations run more smoothly.





# Email Software

One piece of ICT software that you're likely to use regularly is email.

Before the internet and electronic mail, to exchange messages, people had to write letters, and send them through the post. Or, they spoke to each other on the telephone.

As you can imagine, this was time-consuming, and increased the potential for information to get lost.

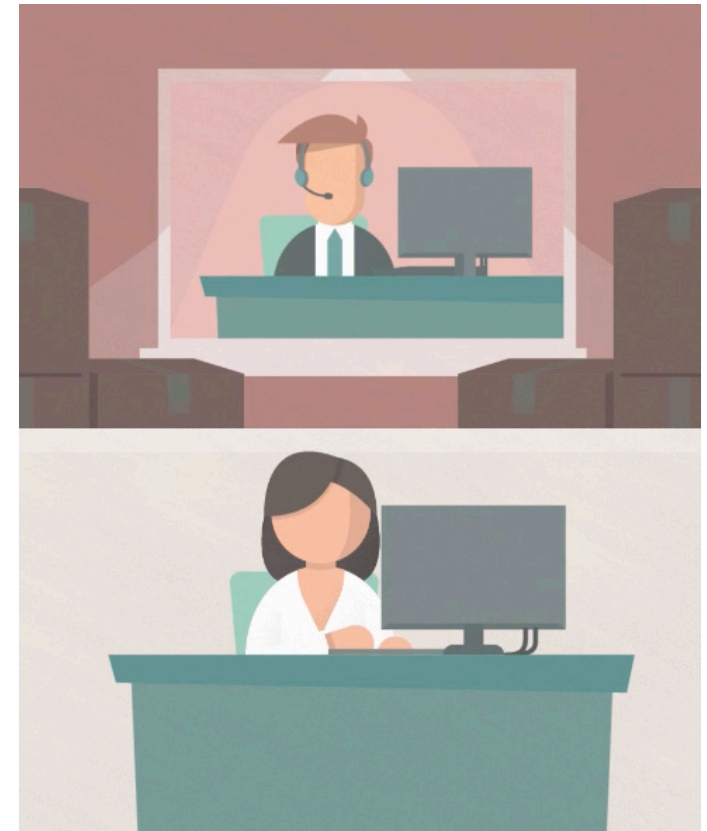


# Efficient Communication

Email software has transformed the way we communicate, and as a result, improved the way organisations across all industries do business.

Its speed means that you can send and receive messages instantly, both within your company and to people outside it – all over the world.

For example, if you received a query from a customer about a missing delivery, you might have to contact the distribution supervisor to find out what's happened before you can reply. Emails make all of this possible within minutes.



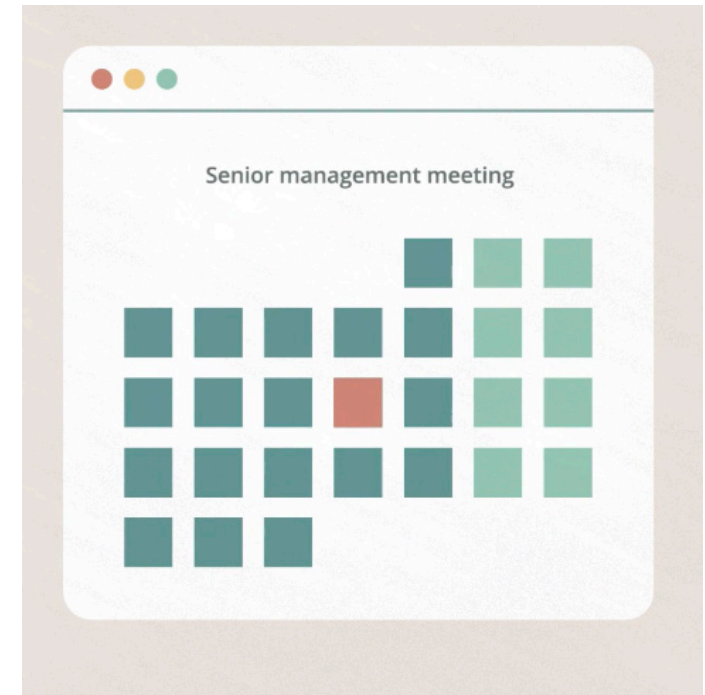


# Calendar and Scheduling

However, email software is a lot more than just a tool for sending messages. Most programs have an inbuilt calendar, which makes the process of scheduling an event or meeting easy. There's also useful functionality to help you organise your daily tasks.

These sections are normally integrated within the software. For instance, if you're asked to arrange a meeting for your organisation's senior management team, you can do so through the email calendar's scheduling functionality.

You would send invitations to invitees by email. And if they accept, you should receive a confirmation notification in turn, and the meeting would automatically be added to their calendars.



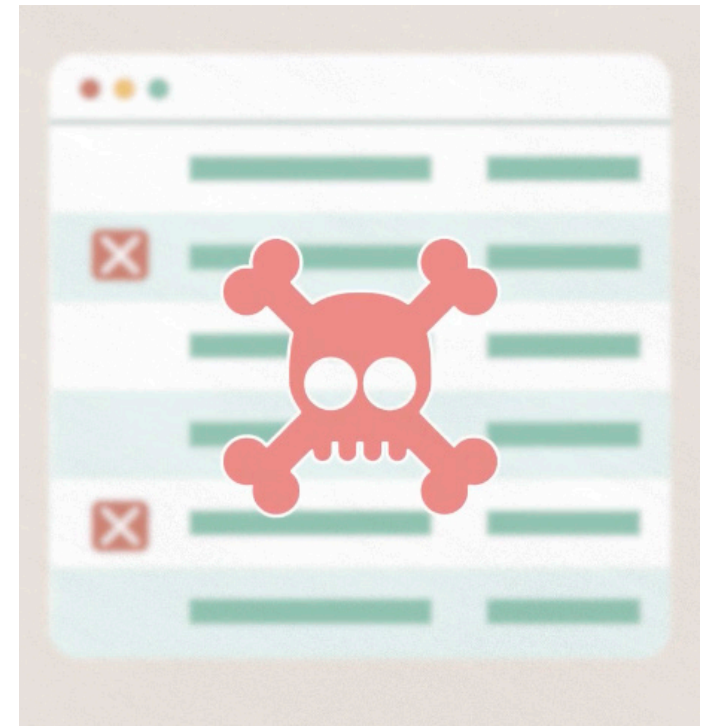


# Streamline Tasks

Email software also has the ability to detect and filter out 'junk' mail. These unwanted messages are not only annoying, but may contain dangerous viruses or corrupt links.

And, when you're busy or on leave, email software can send an automatic message to let people know that you are unable to respond.

Finally, you can also organise any messages you receive into folders and categories, while flagging those that need urgent attention.

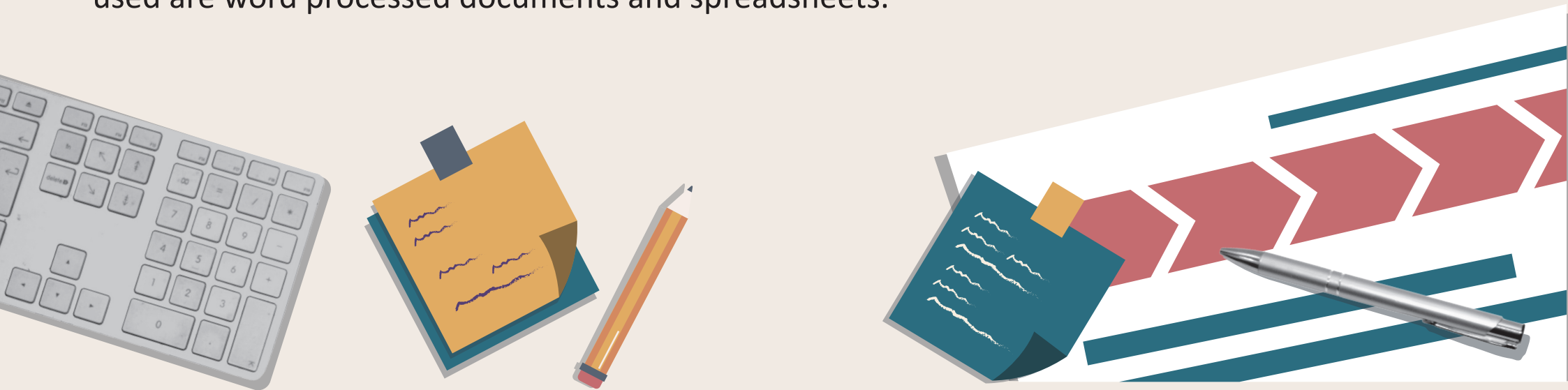




# Attach Documents

One important feature of email software is the ability to send documents as attachments to messages and in some cases, to be reminded if you've forgotten to do so – a fairly common event in business.

These can be in a variety of formats, but two of the most frequently used are word processed documents and spreadsheets.





# Word Processing Software

Word processing software is another major form of information and communication technology. It's used to create, edit, save and print text-based documents.

You'll find it across all business sectors, being used to produce letters, contracts, reports, manuals and articles – to name just a few.

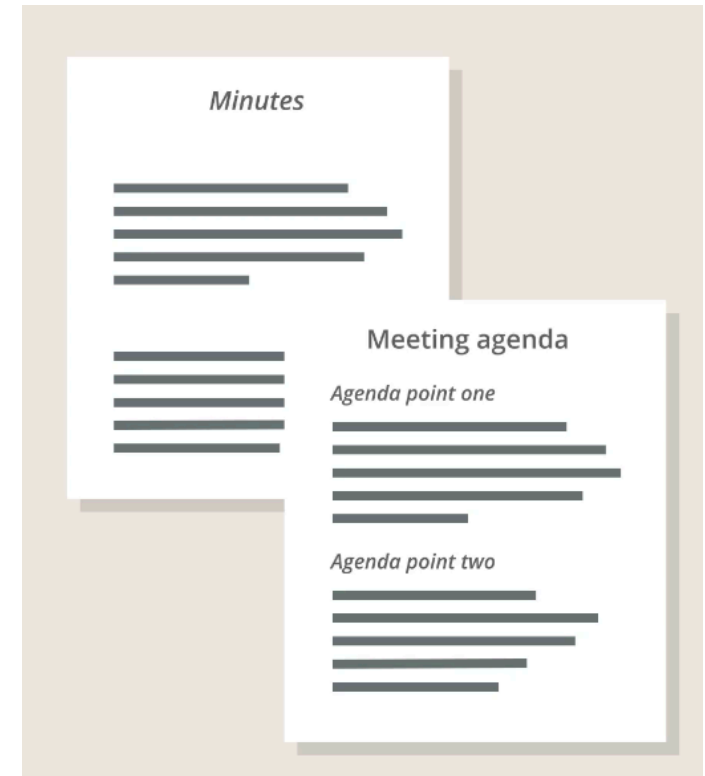


# Manipulate and Edit Text

One key feature of word processing software is its ability to manipulate text.

This can mean copying and pasting it from one place to another, deleting or adding words and letters, and altering how it looks.

For example, when creating the minutes for a meeting, you might build the structure by copying phrases from the meeting agenda, and pasting them into your document.





# Structure and Layout

You could then create headings and bullet points, change the text size, and decide to 'bold' or 'underline' various elements.

You might also change the layout, page orientation and margin sizes. The software will also highlight and offer corrections for spelling and grammar errors.

All of this serves to make a document easier to digest and follow. If you had to write the minutes by hand, you'd need to be sure of your wording before you began, and any mistakes would mean you'd have to start again. You might also have to make copies to distribute, rather than attaching the document to an email.





# Graphics and Images

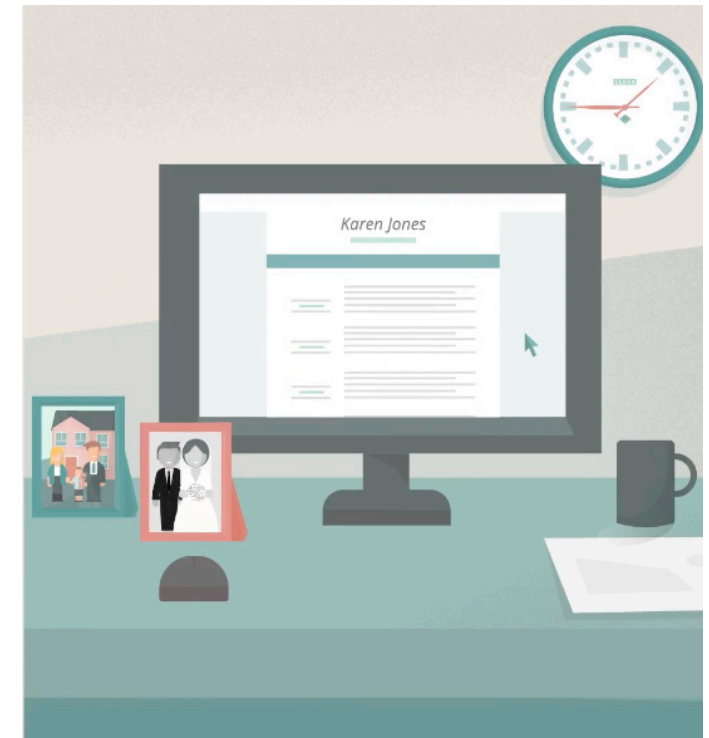
However, word processing software isn't just for text. For instance, if you were creating a report, it would allow you to add graphics alongside the text, as well as images and tables, to enhance visual appeal and illustrate your points.



# Save Progress

And word processing is flexible, in that it can save your work as you go, so you don't lose it. This becomes particularly useful when you're working on things you might return to several times, such as job applications.

As you progress in your career, you can keep updating your saved CV document to reflect your growing achievements and experience, rather than having to start from scratch.



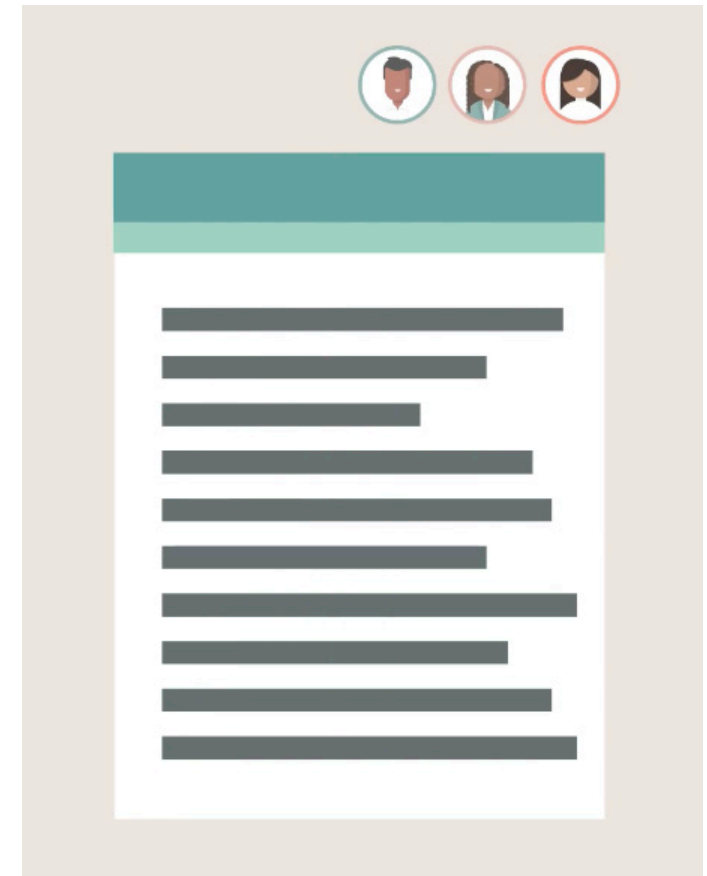


# Collaboration and Review

Word processing becomes even more beneficial when you use web-based programs, because it enables even easier collaboration with colleagues – both inside and outside your organisation.

Documents are stored online and can be accessed by multiple users at once, as long as they have an internet connection and the necessary permissions.

Users can track their changes, ‘tag’ one another in comments, and review previous versions. All of this is particularly useful if you and your colleagues work remotely.



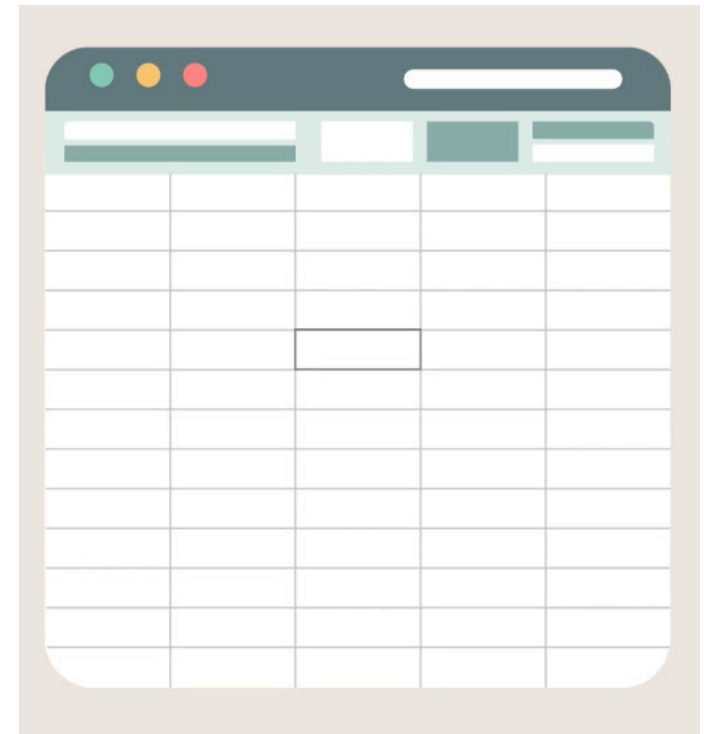


# Spreadsheet Software

While word processing software is typically used for text-focused documents, its equivalent for numbers is spreadsheet software. This is mainly used for arranging, analysing and storing numerical data.

Spreadsheet software originated in the financial sector and is still widespread in financial management and accounting. But it's relevant for almost all business processes – from tracking budgets, to predicting future growth, to creating charts that represent information visually.

The data in a spreadsheet is contained within 'cells', which are created by a grid of rows and columns. It's manipulated and analysed through functions and formulas.



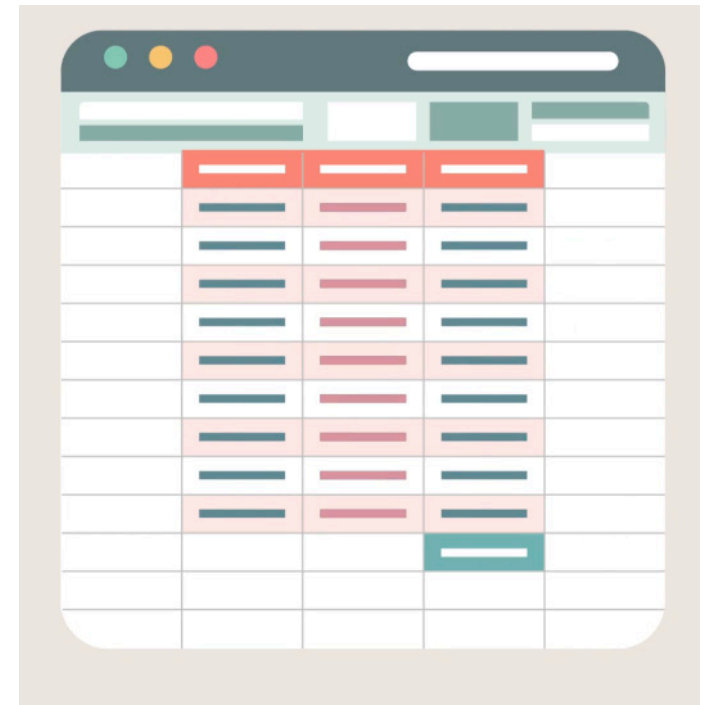


# Calculations, Tracking and Sorting

You may use spreadsheets at work for anything from straightforward calculations of your working hours, to analysing complex customer data.

It might be that you're asked to make a spreadsheet to track product deliveries. The software could help you to create tables, drop-down lists, and automatically calculate totals to make this task easier.

You could also sort and filter information, and use different fonts and colours to make it easier to understand.





## Automate and Update

Without the software, you'd have to do calculations by hand and analyse each piece of data individually, making the task both time-consuming and difficult.

And, just like with word processed documents, the spreadsheets serve as a record that can be saved and changed as needed later on.

# Word Processing, Spreadsheets and Email at Work

If you take a moment to think about what your working – or personal – life would be like without ICT and how you'd carry out even the smallest of tasks, it's clear that ICT is an essential part of the way businesses are structured and operate.

Word processing and spreadsheet software in particular are used at all levels, and emails are a standard form of communication for everyone, across all industries.





# Presentation Software

As well as email, word processing and spreadsheet programs, another form of ICT that is common across all organisations and industries is presentation software.

You can think of presentations as a form of communication designed to convey key messages clearly and persuasively. They're often used in meetings to propose new strategies, deliver financial updates, or evaluate projects and data.

Typically, presentations aren't an exchange of information. Instead, they focus on engaging an audience, and ensuring they not only hear the message, but also understand and retain it.



# Format and Delivery

Presentation software makes this process more efficient and effective through the use of structured, visually appealing 'slides'. These are like 'pages' of a presentation, and serve to illustrate key points.

Without the software, you'd have to rely solely on speech or printed materials, which could make complex information harder to follow, and your delivery less formal.

As a manager, it's likely that you'll need to both create and deliver presentations as part of your role – whether that's conveying important details to your team, gaining support and buy-in from senior leadership, or persuading potential customers.



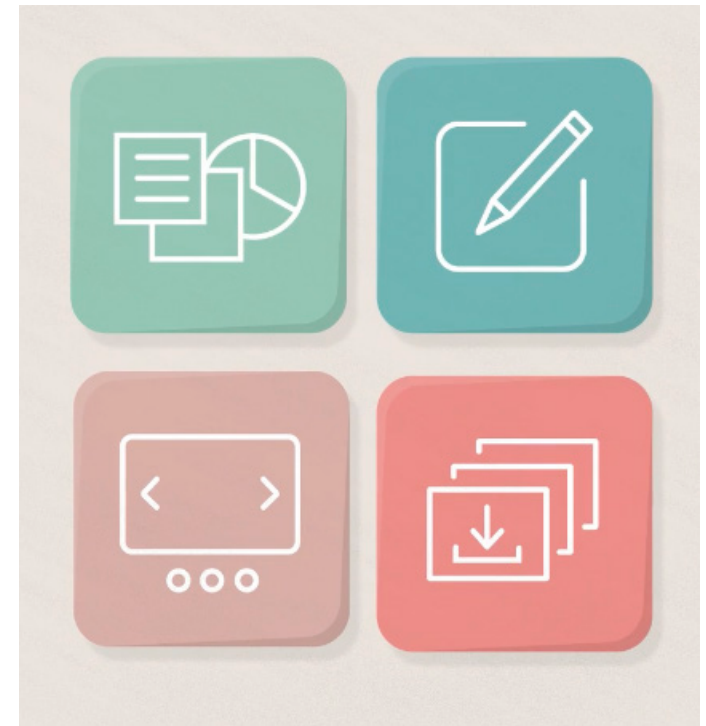


# Slide Decks

You can use presentation software to build, modify, reorder, and save 'decks', or sets of slides, with ease.

There are also useful features, such as templates, which can be used to create a polished and professional look.

Indeed, your organisation may well provide 'master' templates, designed to reflect its corporate brand. This helps to maintain a consistent company identity across all presentations.

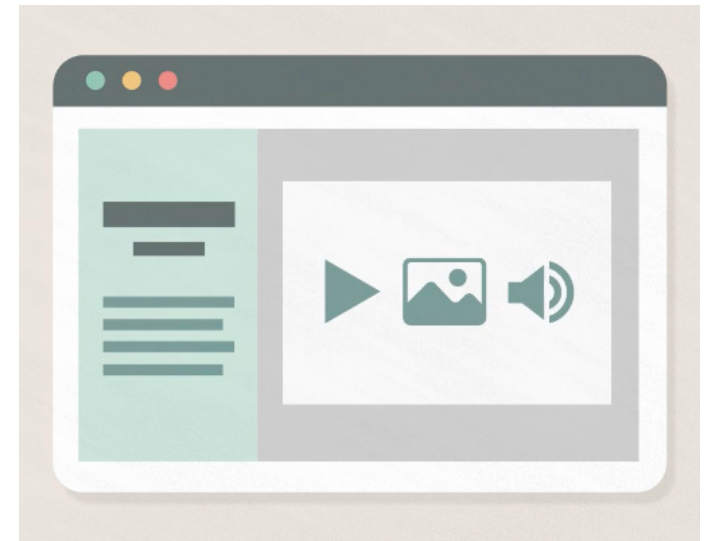




# Multimedia

There are additional features, such as animations – and multimedia elements like videos, images, and audio.

These can keep your audience engaged, and improve their understanding, particularly when presenting data-heavy content. However, it's important to remember that overuse of these elements may have the opposite effect, and lead to distraction and confusion.





# Speaker Notes

Another useful tool is 'speaker notes'. These are added to a presentation but can't be seen by the audience.

They help the speaker to keep track of the main points to cover, without cluttering the slides with too much text.



## Financial and Accounting Software

Let's turn to financial packages, which include accounting software. These systems record business transactions, allowing organisations to meet legal requirements.

They also provide management teams with accurate, real-time financial data, so that they can make well-informed decisions.

Accounting software categorises financial activity to track revenue, expenses, and outstanding payments.

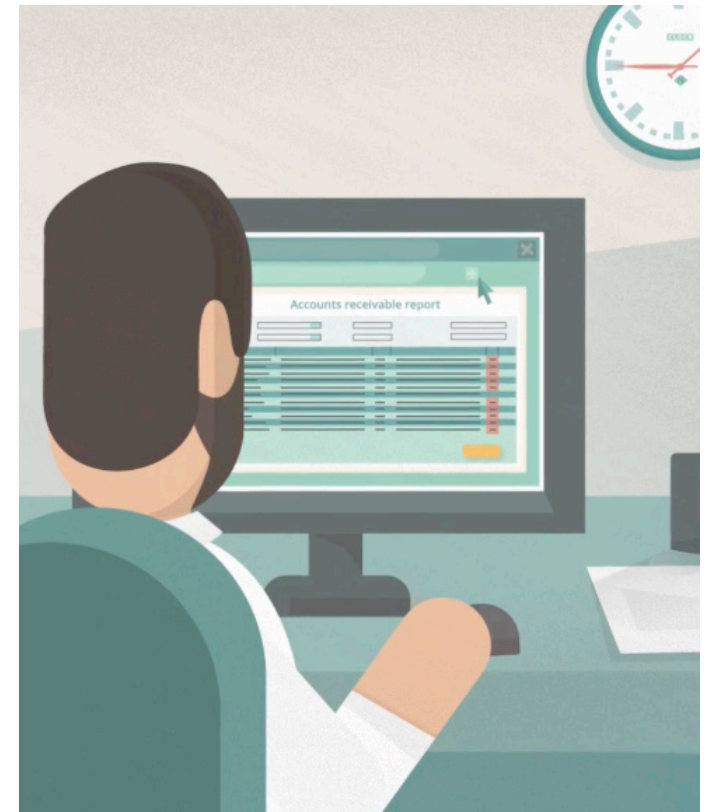


# Budgets, Spending and Payments

As a manager, you may not be directly responsible for operating these systems, but you might use the reports they generate to monitor budgets, forecast spending, or track departmental performance.

For example, you may need to review an 'accounts receivable' report to understand outstanding customer payments.

With accounting software, information like this can be produced at the press of a button, rather than by having to manually check individual transactions.





## Invoices and Purchase Orders

Depending on your role, you might be involved in invoice approvals or purchase order management.

The associated documents can also be generated and recorded in the system and shared digitally, ready to be authorised more quickly.

Accounting software is often linked to an organisation's bank account, so that money received and paid can be matched up with sales and purchase invoices later on.

Without the computerised system to do the work, this task would require a lot more time and effort – not to mention technical skill.



# Off-the-Shelf and Bespoke Packages

Some organisations buy generic – or ‘off-the-shelf’ – packages. Others have bespoke programs designed to suit their needs.

For instance, the NHS or a police force is likely to require custom accounting software, due to their specific requirements.

Meanwhile, a privately owned manufacturing business may be able to use an off-the-shelf product, because it simply needs to track customer and supplier payments.



# Using Bespoke Packages

An off-the-shelf product has been made for the mass market and can be bought and used without the need for customisation. A bespoke system, on the other hand, is specifically designed and created to meet an individual customer's particular needs. In effect, it is tailor-made.

The advantage of bespoke software is that it is custom-built, which means that the recipient organisation can determine how it works. It can be constructed around existing systems in a way that may not always be possible, or as easy, with off-the-shelf software.

Of course, commissioning bespoke software is often significantly more expensive. However, for some organisations it's worth paying for because an effective and efficient system has the potential to deliver competitive business advantages.





# Integration

Whatever its format, accounting software is often integrated with other programs, such as payroll or management information systems – which we'll look at in more detail later.

And while it goes without saying that these tools are used in the financial sector, all businesses will need some form of financial package, regardless of their industry.

In addition to the software we've already discussed, there are other systems that are essential to the effective day-to-day and long-term operations of many companies.



# Customer Relationship Management (CRM) Software

Organisations use customer relationship management (CRM) software to record, track, manage and store information about their interactions with current and potential customers. This means that CRM software is widely used by businesses across many industries.

A CRM system helps staff within an organisation find and share up-to-date information about customers.

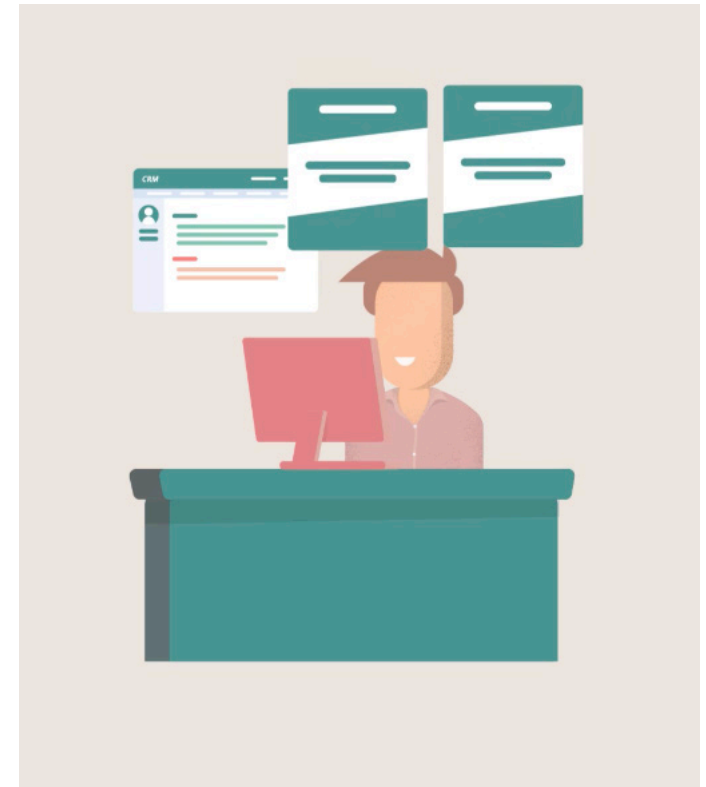
It's particularly useful for people who deal with customers on a daily basis – such as those working in sales, customer services, business development, recruitment and marketing.



# Tracking Customer Interactions

For instance, a call centre representative might add information about an email exchange they've had with a customer to the organisation's CRM.

This data can then be accessed quickly and easily by colleagues across the business. Someone in the sales department could use the insight to finalise a deal, or an employee in marketing might get ideas for a relevant campaign.





## Centrally Shared Information

The idea is that better customer relationships make a business more successful. And that this is based on efficient and effective record management.

Without this data being held centrally, separate pieces of customer information would be kept in different parts of the organisation.

This would make it far harder to access and share, and the business's dealings with customers much less efficient.

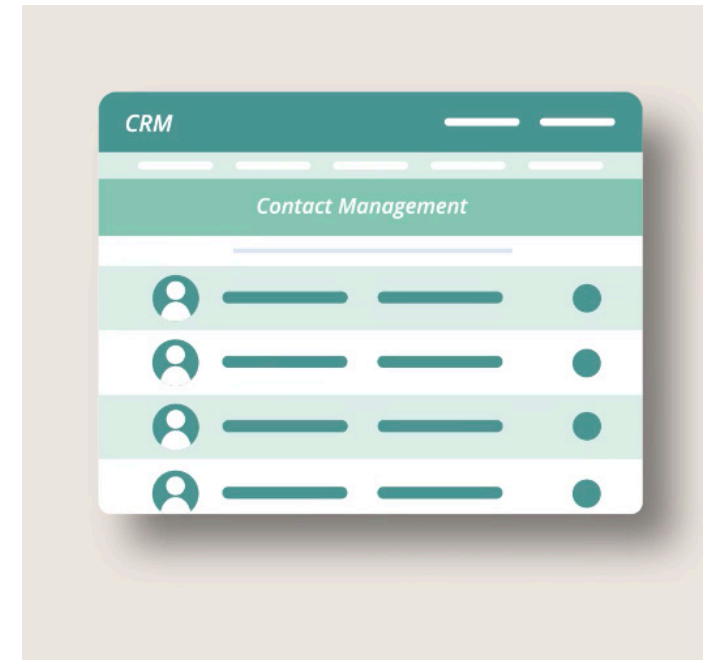


# Contact Management

Like the other types of software, CRM software generally provides an extensive range of features, functionality and tools. Some of these are key to all employees and some are quite specialised – only being used by a small number of people.

However, at its heart, any CRM software should have a 'contact management' feature. This allows users to record factual details about customers, such as names and addresses, as well as make notes about interactions with them.

In this way, the software creates client profiles that work with 'search functionality'. Employees can then easily find and make amendments to records, or add relevant documents, such as quotes and proposals.

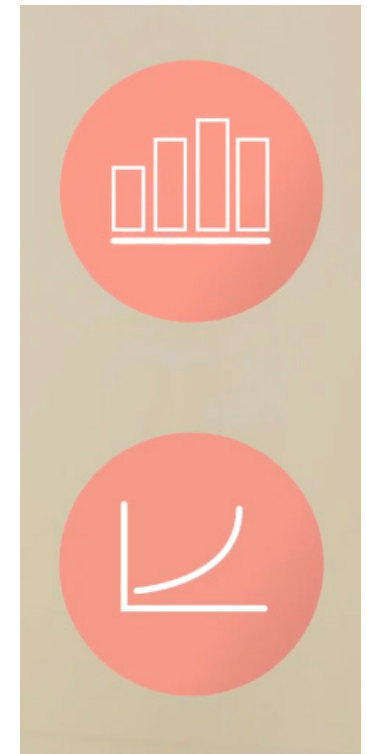




# Communication, Reports and Analysis

Online CRM systems might also integrate with email software, which means that you can send things like tailored marketing promotions from them automatically.

CRM software may also generate reports and analysis, which provide valuable insights into customers' experiences. They also allow the organisation to monitor its sales and performance, and make predictions.



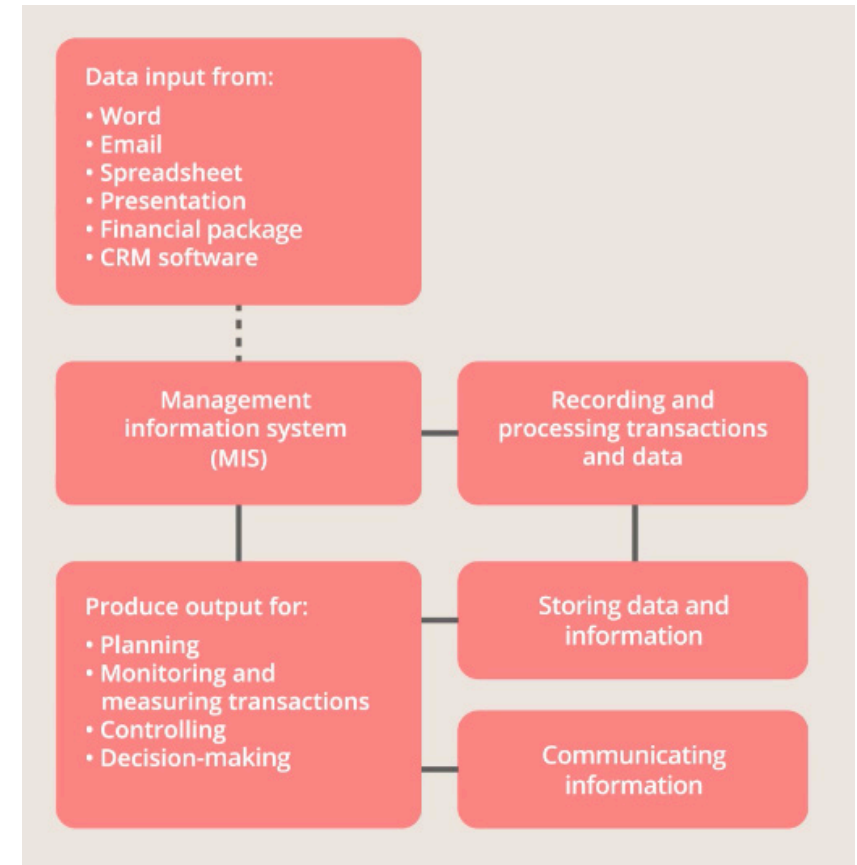


# Management Information System (MIS)

Some organisations may use a standalone CRM system, but others might integrate theirs into a wider database in the form of a management information system.

A management information system (MIS) can be thought of as a hub that gathers data from several ICT packages and manages it in one place.

Its purpose is to turn large volumes of often complex data into usable information for business owners and managers. This aids their planning and decision-making. A CRM is, in fact, an example of a management information system.





## Record Management and Operations

MIS software manages records from multiple sources to produce comprehensive reports, as well as allows different ICT packages to 'speak' to each other.

Through this, it increases an organisation's efficiency by providing accurate, timely and relevant information for staff at all levels, which allows them to make well-informed decisions.

These could be related to daily operational choices, such as ordering more stock, or long-term strategic decisions. For example, about expansion plans or investing in new equipment or vehicles.

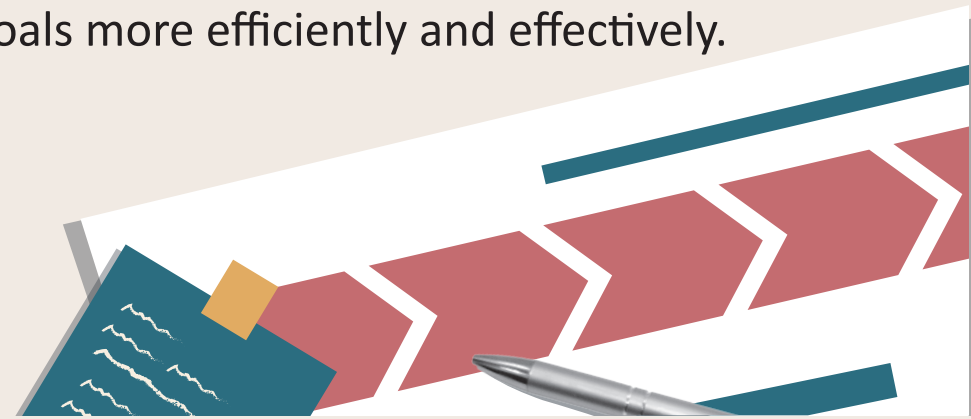
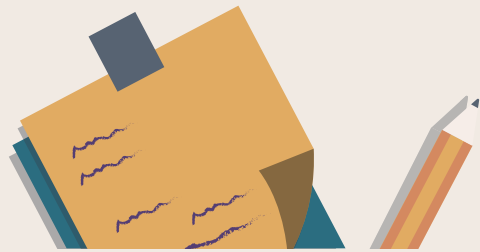
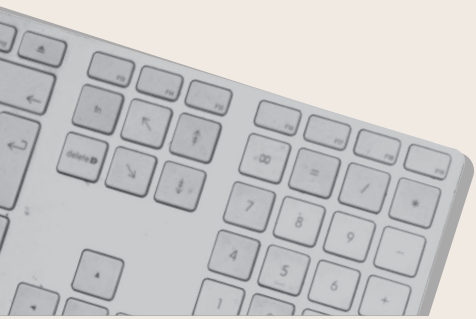


# Broader Role of Technology

The number and variety of ICT business platforms and packages an organisation uses will vary depending on factors such as the sector it operates in, its size and the complexity of its operations.

But technology in business extends beyond just the tools we use – it plays a crucial role in supporting overall operations, such as solving problems, facilitating collaboration, managing transactions and ensuring security.

As such, it's important not to view technology as an end in itself – rather as a set of tools that enable managers and their teams to achieve business goals more efficiently and effectively.



# Digital Skills Framework

However, with nearly a quarter of the workforce lacking the necessary digital skills for work, the UK government has put together an 'essential digital skills framework' that outlines the fundamental competencies needed for individuals to thrive in the modern workplace.

Building on this, Skills England has designed a framework of five digital skills and accompanying characteristics to support current – and future – employees:

- Problem-solving
- Digital collaboration and communication
- Facilitating digital transactions
- Organisational security
- Handling data securely





# Problem-solving

Problem-solving in the workplace increasingly relies on digital tools to help managers and teams diagnose and analyse issues – as well as find appropriate solutions.

This might involve sorting out problems with the technology itself – recognising common issues, performing fault-finding, and using online tutorials to fix errors.

Being able to do this for yourself can make processes more efficient, and free up your organisation's IT support team to focus on more complex challenges.

At the same time, it's important to recognise the limits of your technical expertise, and know when to escalate difficulties.





# Research and Analysis

Digital problem-solving skills are also important when it comes to conducting research, because businesses increasingly rely on data-driven insights to guide decision-making.

Managers must be able to access, analyse and interpret large amounts of data from a range of sources. This includes knowing how to use search engines to get reliable information, as well as navigate databases and extract relevant details.

Data analysis might also involve using specialised market research platforms, online forums, and competitive analysis tools to monitor trends and build up background industry knowledge.



## Data Visualisation Tools

Linked to this is the ability to use data visualisation tools to manipulate and interpret complex data sets, identify patterns, and share findings.

This makes problem-solving more efficient, and improves the communication of information.



# Accuracy, Reliability and Efficiency

When sourcing and accessing data for problem-solving, it's important to verify that it's accurate, reliable, and free from errors.

You should also be mindful of potential biases that could distort findings and result in misleading conclusions.

And, when generating solutions to problems, you should look for opportunities to automate processes and tasks to improve efficiency – while ensuring that all methods are legal and ethical.



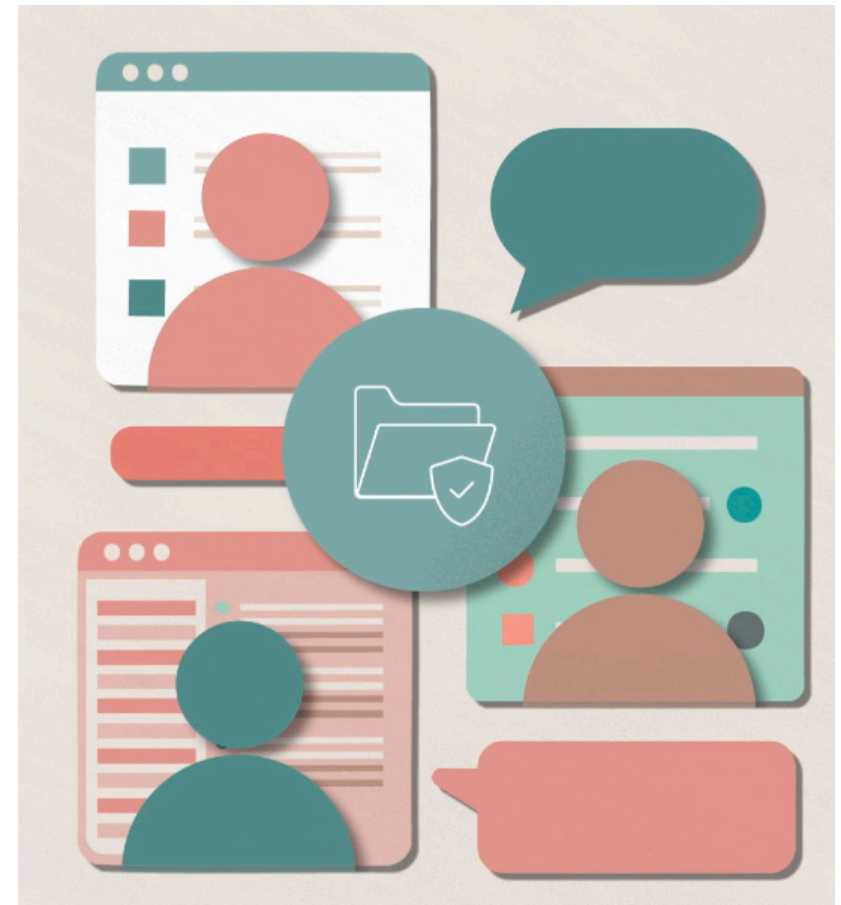


# Digital Collaboration and Communication

Let's now look at the second key skill – 'digital collaboration and communication'. This isn't just about sending emails, it means understanding and being able to use tools to exchange ideas and synthesise knowledge.

Indeed, effective digital communication involves a range of collaboration platforms, which facilitate efficiencies such as real-time messaging, video conferencing, and file sharing to manage tasks.

It's also important to choose the correct communication method. For example, while instant messaging is useful for quick updates, complex discussions may require a video call to enable interaction and a deeper exchange of ideas.





# Storing Files and Working Simultaneously

Elsewhere, 'cloud-based' document management systems allow for centralised storage and version control, file sharing, and real-time collaboration.

Team members can work on a document simultaneously, to edit, review and comment, without the need for additional communication, which makes carrying out tasks more efficient.





## Accessing Training and Development

Managers should also promote the use of digital tools and platforms to support team members' personal development and upskilling.

Traditionally, many organisations have favoured face-to-face training courses. However, digital tools have made learning more accessible, flexible and scalable.

This allows employees to upskill through online courses, collaborative virtual workshops, webinars, podcasts and interactive learning platforms – at a time and pace that suits them.



# Digital Transactions

The third key skill is ‘facilitating digital transactions’. Skills England defines these as “online or automated transactions that take place between people and organisations – without the use of paper, and typically using digital business systems”.

One example is CRM systems. Organisations use these to store and process customer information. Managers should understand how CRMs facilitate sales, marketing and customer service processes, in compliance with data protection regulations.



## Managing Records and Forms

You must also ensure that you know how to access and manage digital employee records for your team members. This is often done through an MIS.

As a manager, you might use an MIS for handling leave requests, recording absence, updating employee records, and storing performance reviews.

It's also important to be familiar with the tools your company uses to submit digital forms, such as those for supplier contracts or expenses. Managers need to make sure that both they and their teams can complete, verify and submit these documents securely and accurately.



## Digital Skills for Security

The final two skills in Skills England's digital skills framework – 'organisational security' and 'handling data securely' – focus on safeguarding business operations.

For managers and their teams, organisational security – or cybersecurity – and secure data handling are key to protecting the business as well as its employees and customers.

Skills England defines 'organisational security' as "a process to create rules and actions to take to protect individuals and organisations against cyber attacks". Essentially, organisational security protects systems and networks from such harms.

Meanwhile, 'handling data securely' is more concerned with making sure that data itself is protected from unauthorised access, modification, damage or deletion. In practice, however, there is much crossover between the two.





## Organisational Security

Cyber threats – such as ‘malware’ and ‘phishing’ – pose significant risks to organisations. It’s therefore essential for companies to implement robust security measures.

The National Cyber Security Centre suggests five simple steps for improving cyber security:

- Backing up your data
- Protecting your organisation from malware
- Keeping smartphones safe
- Using passwords to protect data
- Avoiding ‘phishing’ attacks

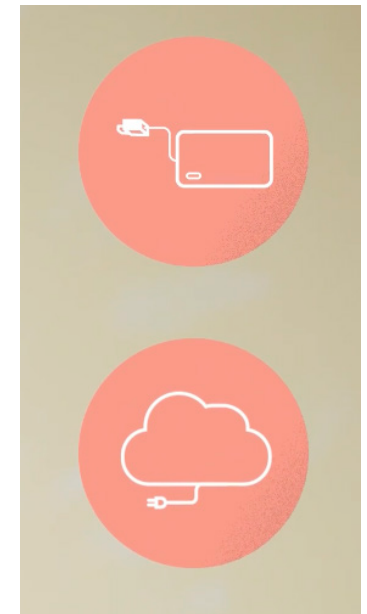


# Back Up Data

First, backing up data ensures that in the event of damage or theft, an organisation can recover important files. If a company can restore access themselves, it also means they can't be blackmailed by 'ransomware' attacks.

Backing up data involves saving a copy to another device – for example, an external hard drive, a networked server, or an online cloud storage provider.

The National Cyber Security Centre states that “all businesses, regardless of size, should take regular backups of their important data, and make sure that these backups are recent and can be restored”.



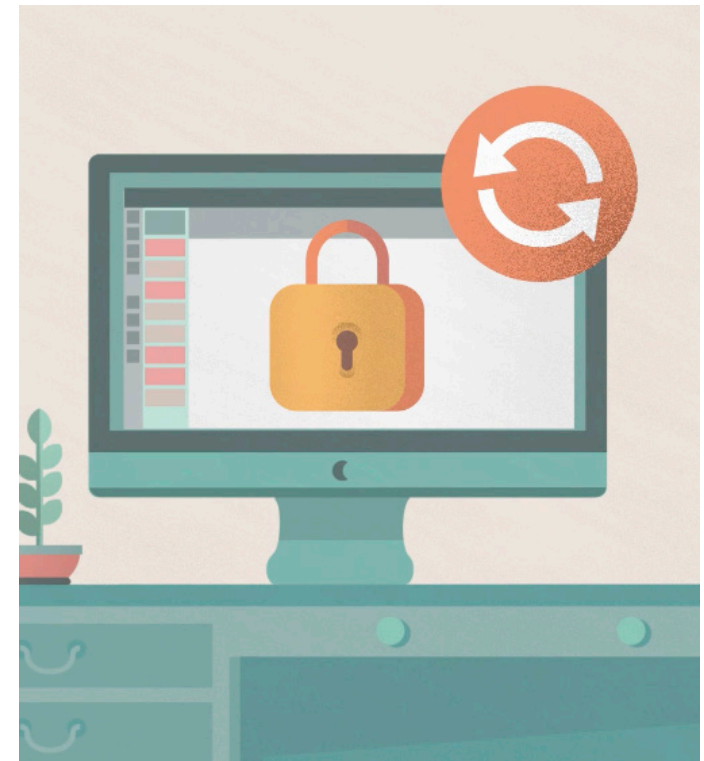


# Protect Against Malware

Second, protecting your organisation from malware – or ‘malicious software’ – involves installing an antivirus program, updating it regularly, and educating staff on safe practices.

This reduces the risk of viruses and other harmful network disruptions. All IT equipment – laptops, PCs, tablets and smartphones – should have the latest software and firmware installed.

Enabling automatic updates ensures that critical security ‘patches’ are applied, which strengthens protection against threats.





## Keep Smartphones Safe

Third, keeping smartphones safe means making sure that any mobile devices people use for work are protected by strong passwords, encryption and 'remote wipe capabilities'.

Smartphones can be an easy target for hackers, so taking steps to protect them – and educating employees on these measures – is essential.



## Use Passwords

Fourth, using passwords to protect data means that only authorised people can access sensitive information.

Accounts should have strong, unique passwords and multi-step verification. This asks for more information to prove identity – such as a code sent to a separate, authorised device.

So, even if hackers manage to obtain the password, they'll be unable to access the account.



# Avoid Phishing Attacks

Lastly, avoiding phishing attacks involves training employees to recognise suspicious emails or texts, and to not click on fraudulent links, or potentially harmful attachments.

Your team members should also be made aware of ‘spear phishing’, so-called because it targets a particular person or organisation. It uses actual names, job roles or social media profiles to make the messages more believable.

Giving employees regular cybersecurity reminders can help to reinforce good habits. For instance, organisations might display password security tips on login screens, or provide phishing awareness updates on the company intranet.



## Handling Data Securely

Much of this organisational security advice also applies to handling data securely.

For example, strong passwords and multi-step verification can also help to safeguard against data breaches and leaks.

However, handling data securely is also about how data is collected, stored and shared. This includes potentially sensitive personal data and information – for instance, customers' names and addresses, and your team members' appraisal records.



## Collection and Storage

When collecting data, managers and employees must ensure that it's gathered legally and ethically, with proper consent from individuals if required.

For storage, personal and sensitive information should be encrypted, well organised, and kept in secure locations. Managers and their teams must adhere to strict access controls, ensuring only authorised individuals can view or modify data.

This is particularly crucial when using cloud software, because these environments often involve multiple users, devices and services, which increases the number of potential entry points for threats.



## Sharing and Distribution

Finally, data should be shared with caution, so that only authorised individuals receive it.

And distribution must be via secure channels, such as encrypted emails or protected file transfer systems.



## Insider Cybersecurity Threats

The practices we've outlined help to reduce the risk of insider threats – cybersecurity risks that come from within an organisation.

Insider threats can be either negligent – as in the case of an employee clicking on a suspicious link or malicious – where somebody within, or connected to, the organisation deliberately tries to carry out a cyber attack.



# Strong 'Security Culture'

By understanding organisational security and ensuring that all team members handle data securely, line managers can help to protect their teams, customers, and the business from digital threats.

A strong 'security culture', backed by practical measures such as regular training and vulnerability assessments, prepares both employees and systems to handle threats.

Much of this preparation is highly relevant to daily and ongoing organisational operations. However, the nature of technology means that it is always developing and changing.



# Future of Technology in Business

But what might the future of technology mean for business?

According to Jeremias Adams-Prassl, Professor of Law, University of Oxford, “there is no such thing as *the* future of work”.

In other words, there is not one, pre-determined way in which organisations will operate in the future. From our perspective in the present, we can’t know what the future will look like, or exactly what technology’s place in the world of work will be.



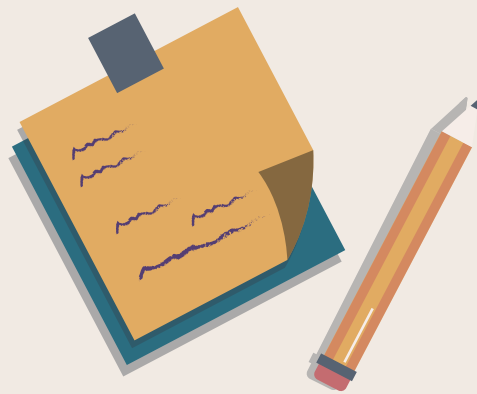
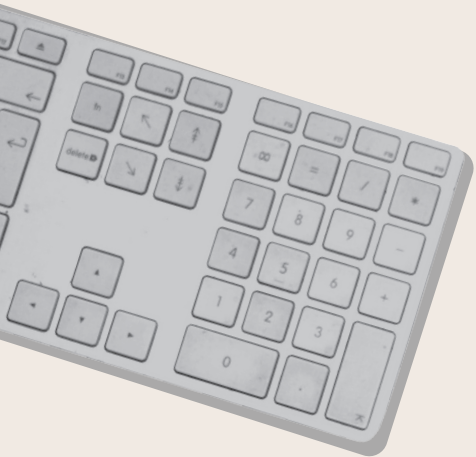


# Adoption of New Technology in Organisations

There are an infinite number of possibilities that could become reality – and our decisions today can, and will, influence the working lives of tomorrow.

It is therefore vitally important that organisations and managers understand how decisions they make now about the use of technology could affect their employees and their business.

Let's explore the impact of technology adoption in organisations.





# Effect on Employees

As organisations adopt new technologies, it's vital that they understand how their decisions are likely to affect employees.

With this in mind, the Institute for the Future of Work, an independent research and development body, has identified six 'impacts' of technology adoption that organisations should consider.

We'll examine each in turn.



# Substitution

First we have 'substitution', in which technology carries out new tasks, or even whole roles, that were previously performed by people.

For instance, replacing basic customer service activities with a 'chatbot'.

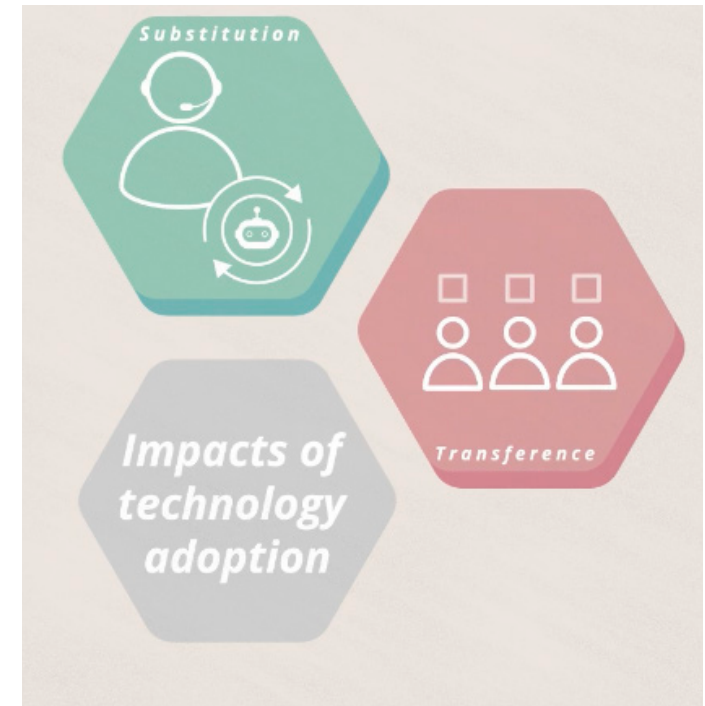




# Transference

Next is 'transference', where technology allows ownership of a task to be 'transferred' from one person to others.

An example might be employees booking annual leave through a self-service platform, rather than their manager or HR department logging it for them.





# Telepresence

The third impact is 'telepresence', or technology that allows people to work from any location.

It includes collaborative cloud-based software, instant-messaging platforms, and video-conferencing programs.





# Creation

Next we have 'creation' – when technology produces new tasks or jobs.

For instance, the advent of artificial intelligence (AI) has led to a range of additional roles, including 'AI ethicists', who establish best-practice guidelines and standards for responsible AI development.





# Augmentation

The fifth impact is 'augmentation', which is about increasing employee capabilities through technology.

An example might be architects using virtual reality headsets. They can simulate being inside a building before construction, and identify potential problems or refinements.





# Intensification

Lastly, we have ‘intensification’, which looks at efficiency – doing more tasks in the time available.

For instance, warehouse workers using robots and automated systems to pick multiple orders simultaneously.





# ‘Low-road’ and ‘High-road’ Approaches

The Institute for the Future of Work has proposed that organisations deciding on new technology should carefully consider these impacts in relation to employees. A company’s actions in this area will affect whether it takes a ‘low-road’ or ‘high-road’ approach to technology adoption.

The ‘low-road’ approach focuses on short-term economic gains at the expense of human experience. The result of this might be low-quality work, which is closely linked with decreased productivity, and negative consequences for employees and society.

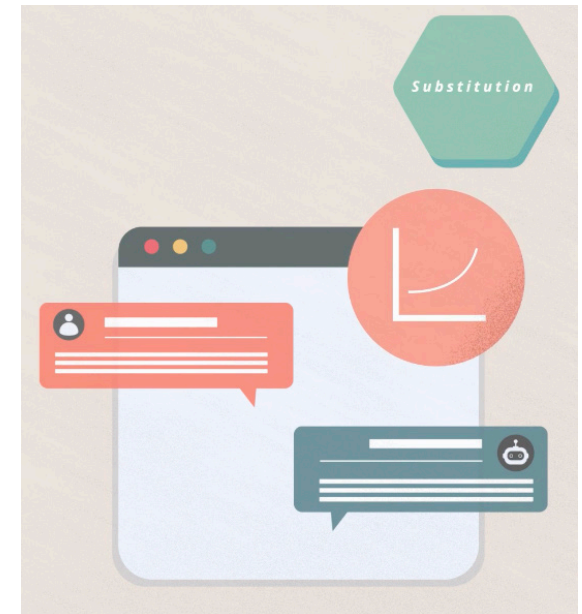
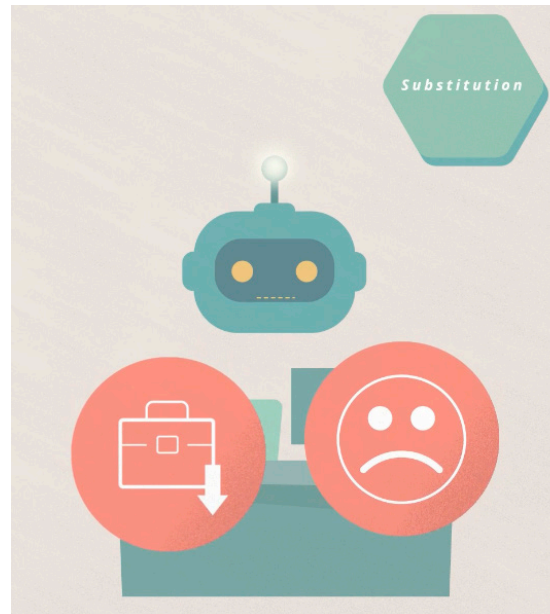
Conversely, the ‘high-road’ approach values good work practices like fair pay and conditions, equality, dignity, and autonomy. Embracing these principles can lead to greater employee wellbeing, which has in turn been shown to improve long-term productivity.



# Low-road and High-road in Practice

For example, a 'substitution' could be made to replace human-led customer service tasks with chatbots. If this is done purely to cut costs, it may result in job losses, and a decrease in morale and wellbeing. This would be a 'low-road' approach.

However, if the technology is focused on automating tedious tasks, which frees up employees to do more meaningful, value-added work, it may lead to greater engagement and productivity. This represents a 'high-road' approach.

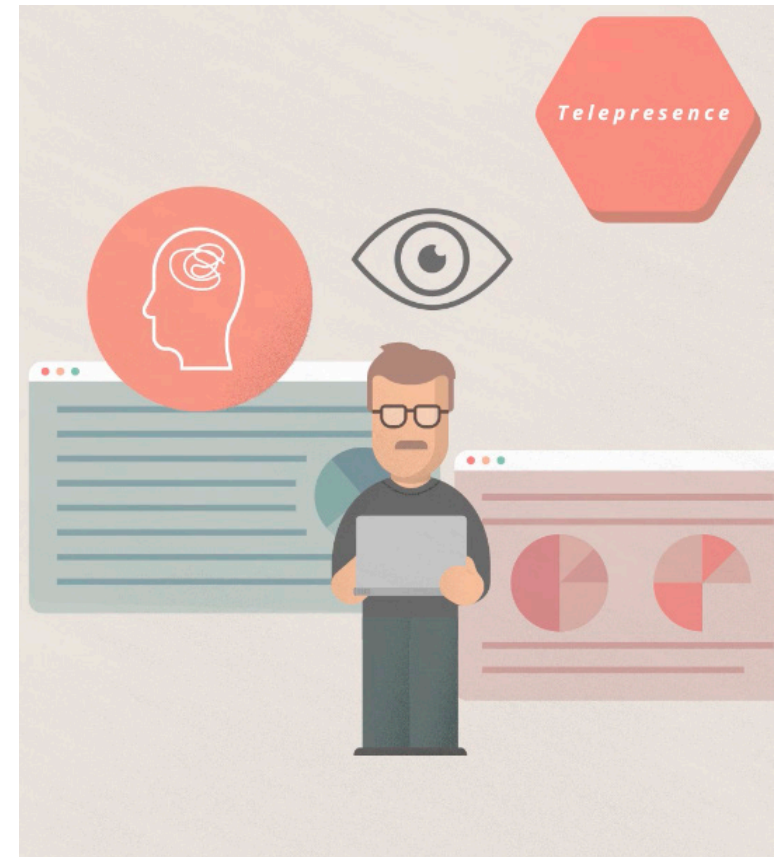




# Low-road and High-road in Practice (cont.)

Similarly, an organisation's attitude towards 'telepresence' can have a significant positive – or negative – impact. If managers use remote monitoring tools to micromanage team members, and demand that they're constantly available, it may lead to stress and burnout – again, taking the 'low road'.

In contrast, companies can use 'telepresence' to offer employees flexibility and autonomy – while maintaining effective communication. This may foster trust, work–life balance, and job satisfaction, which would be the 'high road'.





# Manager's Role in Technology Adoption

Organisations often face multiple technological impacts at the same time – so it's important to consider how they interact and influence both immediate and long-term outcomes for the workforce.

As a manager, you have a key part to play in enabling and developing your team to get the most from the adoption of new technologies.



## Complement with ‘Human’ Skills

One challenge here is the understandable fear that new technology – particularly AI – will replace human jobs. However, as we’ve discussed, automating routine or mundane tasks can allow people to focus on more impactful, interesting and rewarding work.

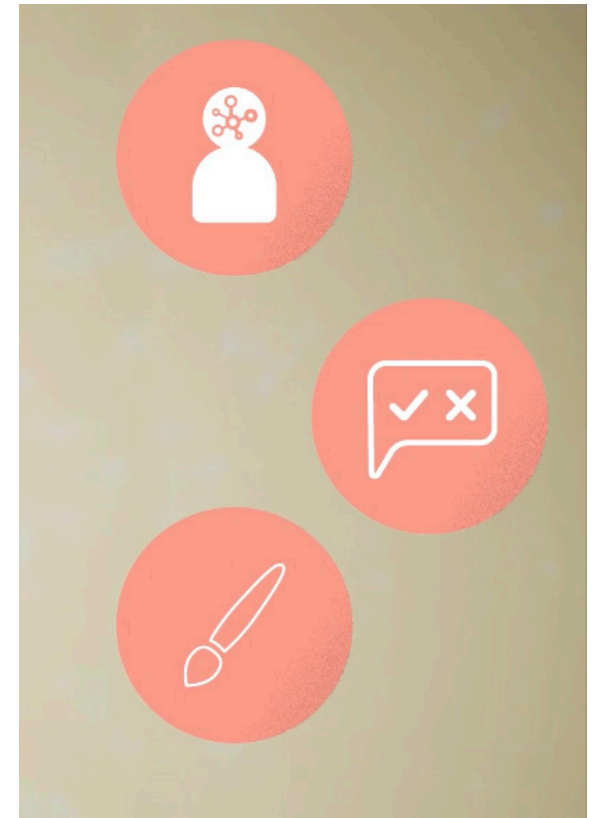
You can help here by developing your team’s higher-level ‘human’ skills to complement technology – such as critical thinking, decision-making, and creativity.



## Experimentation, Learning and Development

It's also useful to foster a culture of experimentation, shared learning, and continuous development.

And you should encourage your team members to keep themselves informed about new advances in technology – offering them training and personal development in this area where you can.



# New Technologies

We've looked at how organisations need to think carefully about the short-term and long-term impacts of adopting new technology and the effect it can have on people.

But which technologies are likely to influence how we work in the future? As we've seen, it's impossible to predict what the future will look like with any certainty – but what we can do is be aware of the technologies that are changing the way we work today and use them as a guide about the likely direction of travel.



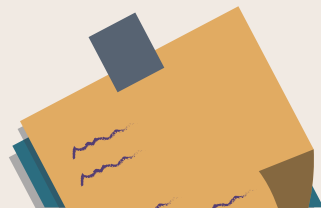


# Automation

Automation and artificial intelligence (AI) are related concepts that aim to make work processes more efficient and effective.

Automation describes “any technology that reduces human labour, especially for predictable or routine tasks”. For example, at home, most of us use a washing machine to automate laundry. In the workplace, this might include robots on assembly lines or software that automates data entry.

While traditional automation – also known as ‘robotic process automation’ (RPA) – follows preset instructions, doing the same task or set of tasks consistently, AI may be also used to enhance automation.



# Artificial Intelligence (AI)

AI has no single agreed definition but you can think of it broadly as the simulation of human abilities, skills and behaviours by computer systems.

There are a wide range of disciplines sitting under the AI umbrella that address different aspects of mimicking human intelligence. These overlap and are often used in combination.



# Machine Learning and Generative AI

For instance, machine learning simulates human learning – computer systems learn from data to improve their accuracy and performance without being programmed with specific rules. If you’ve ever been recommended a song by your music streaming service based on your previous listening, you’re experiencing machine learning in action.

Another form of AI that you may well be familiar with is ‘generative AI’. This mimics the human ability to be creative and generate new content – including images, audio and music or text. For example, there are a number of tools that can be used to respond intelligently to human input or questions and generate original human-like text.

Discover more about automation and AI and how they’re impacting the world of work by downloading this lesson’s associated information sheet.





# Recap

In this lesson, you have learned about:

- Role of technology and its applications in the workplace
- Using business software packages to produce documentation, optimise day-to-day tasks and support communication
- Key digital skills
- Future of technology – key trends and opportunities